REHOTEQ

AI EDUCATION CURRICULUM

Inter-School Coding Competition Pack

Junior Secondary (JSS 1 to JSS 3)

Field	Detail
Document type	Competition organisation pack: rules, rounds, scoring rubric, sample challenges and certificates
Designed for	Host schools running an inter-school coding competition for Junior Secondary pupils
Companion materials	Rehoteq AI Education Curriculum lesson notes (JSS 1, First Term, Weeks 1 to 10) and printable certificate PDFs
Academic session	2026/2027
Curriculum status	Proposed materials for school pilot, review and consultation
Prepared by	Rehoteq Technologies (rehoteq.com)
Edition	First edition, November 2026

PROPOSED MATERIALS NOTICE: This pack expands the Rehoteq AI Education Curriculum, a proposed curriculum for school pilot, review and consultation. It is not represented as an officially approved national syllabus and does not replace the current NERDC curriculum or any examination-board requirement. Host schools remain responsible for their own safeguarding, consent and examination-calendar arrangements.

1. About the Competition

The Rehoteq Inter-School Coding Competition is a one-day (or one-morning) event that lets Junior Secondary pupils show what they learned from the Rehoteq AI Education Curriculum: computational thinking, unplugged problem solving, basic web pages, and simple Scratch-style or Python logic. It is designed for real Nigerian school conditions.

- Low-tech by design: Round 1 needs only paper and chalk. Rounds 2 and 3 can run on as few as one shared computer per team, and every practical challenge has an unplugged fallback so no school is locked out.
- Competition questions are drawn from the same scheme of work the pupils were taught, so the event rewards classroom learning rather than private tutoring.
- Teams compete for their school, which builds school pride and gives teachers a showcase for the curriculum.

Health, safety and consent are the host school's responsibility. Every visiting pupil must arrive with a teacher or approved guardian, and the host keeps a signed consent record for the day.

2. Who Can Enter

Item	Rule
Classes	JSS 1, JSS 2 and JSS 3 pupils of participating schools
Team size	3 pupils per team (schools may bring one reserve, who may substitute before Round 2 begins)
Teams per school	Up to 3 teams per school
Supervision	At least one adult supervisor per school, present for the whole event
Entry fee	Set by the host school (a modest token to cover materials is normal; no pupil should be excluded for inability to pay)
Registration	Schools register teams with the host school at least 7 days before the event, naming each pupil and their class

3. Competition Format: Three Rounds

A full event fits a school morning (about 8:30 AM to 1:00 PM). Where computer access is very limited, Rounds 2 and 3 can run in the unplugged fallback mode described in Section 6.

Round	Name	Time	What happens	Needs
1	Unplugged Quiz	45 min	Mixed teams answer questions on the day's topics: parts of a computer, how the internet works, what AI is and does, logic and patterns. Written answers, one answer	Paper only

			sheet per team.	
2	Build Round	75 min	Teams build a working solution: a Scratch-style program, an HTML page or a Python script, on the challenge drawn for their level. Judged live against the rubric.	1 computer per team (or unplugged fallback)
3	Project Showcase	45 min	Each team presents its build in 3 minutes: what it does, how it works, one limitation and one improvement they would make next.	Poster or screen; no projector needed

Suggested running order

1. 8:30 Arrival, registration, seating. 8:45 Welcome and rules briefing (10 minutes).
2. 9:00 Round 1: Unplugged Quiz (45 minutes), while judges prepare.
3. 10:00 Break (20 minutes).
4. 10:20 Round 2: Build Round (75 minutes).
5. 11:40 Short break while judges score.
6. 11:55 Round 3: Project Showcase (45 minutes).
7. 12:45 Scores tallied, certificates and prizes presented, closing remarks.

4. Scoring Rubric

Total score per team: 100 points. Round 1 is marked from answer sheets; Rounds 2 and 3 are judged live by at least two judges, and the two scores are averaged.

Round 1: Unplugged Quiz (30 points)

30 questions, 1 point each. No partial marks. Questions come from the taught scheme of work only.

Round 2: Build Round (50 points)

Criterion	Max points	What full marks look like
Correctness	20	The solution does what the challenge asked, on the judges' test example
Decomposition and logic	12	The task is broken into clear steps; repetition and conditions are used sensibly
Working under constraints	8	The solution handles the challenge's

		stated edge case (for example, an empty input or a wrong value)
Teamwork and time use	5	All three members contributed visibly; the team finished within the time
Explanation quality	5	The team can explain any part of their solution when a judge asks

Round 3: Project Showcase (20 points)

Criterion	Max points	What full marks look like
Clarity	8	A visitor who missed the build understands it from the presentation alone
Honest reflection	6	The team names one real limitation and one sensible improvement
Confidence and teamwork	6	Members share the talking; questions are answered by the member who did that part

Tie-breaker order: (1) higher Round 2 Correctness score, (2) higher Round 1 score, (3) a single live tie-break challenge of 5 minutes judged on Correctness alone.

5. Judging, Fairness and Appeals

- At least two judges per panel; where possible one judge is a teacher from a school with no team in that panel's room.
- Judges score independently on the printed rubric before comparing notes, so the rubric, not the loudest voice, decides.
- Challenge papers are sealed until the round begins and are the same for every team at the same level.
- A school that believes a scoring error occurred may raise it with the chief judge within 15 minutes of scores being posted. The chief judge's written decision is final.
- Any form of outside help during a round (phones, notes, coaching from the audience) earns a warning on the first occasion and disqualification of the team on the second.

6. Sample Challenges

These are practice and calibration examples. The host school sets fresh challenges for the event itself from the same scheme of work. Every challenge carries an unplugged fallback in brackets.

Round 1 sample questions (with answers)

8. Name three input devices and one output device. (Keyboard, mouse, microphone, touchscreen; output: monitor, printer, speaker.)

9. What does the operating system do in one sentence? (It manages the computer's resources and lets programs and users work with the hardware.)
10. Explain to a Primary 6 pupil how an email travels from Lagos to London. (It is broken into packets, sent through internet cables and routers, and reassembled at the destination.)
11. What is artificial intelligence? (Computer programs that learn patterns from many examples and use them to answer, predict or decide.)
12. A bank's app alerts a customer about unusual spending. What pattern did it learn, and who confirms the action? (The customer's normal spending pattern; a human bank officer confirms.)
13. In classification, what must a machine see many of before it can group new things? (Labelled examples.)
14. Sort these into two groups and name each group: bottle top, leaf, tin, seed. (Man-made or non-living: bottle top, tin; natural or from plants: leaf, seed.)
15. Write the steps a robot would follow to make a cup of tea, using a repeat where useful. (Any correct ordered steps; a repeat for stirring is a bonus.)
16. What could go wrong if a machine learns from one-sided data? (It learns wrong or unfair patterns and makes wrong or unfair decisions.)
17. Give one task AI does better than humans and one humans do better, with a reason for each. (Examples: watching thousands of transactions quickly; understanding feelings and making fair final decisions.)

Round 2 sample build challenges

JSS 1 track (Scratch-style or blocks, or paper flowchart fallback):

- "Traffic light helper": make the character cycle through red, yellow and green, waiting 3 seconds at each colour, forever. Extension: honk (play a sound) when the light turns green. [Unplugged fallback: draw the flowchart with loops and conditions.]

JSS 2 track (HTML or blocks):

- "My school page": build a web page with a heading, two paragraphs about your school, one image and a list of three subjects. Extension: a link that jumps to the subjects list. [Unplugged fallback: write the page on paper with correct tags.]

JSS 3 track (Python or any language):

- "Score sorter": read five scores, print the highest, the lowest and the class average. Extension: also print how many scores are at or above 70. [Unplugged fallback: write the pseudocode and trace it by hand on a worked example.]

Round 3 sample project briefs

- Build a small thing that helps your school, present it, and say honestly what it cannot do yet. Examples from the taught scheme: a revision quiz about Week 1 to 10 topics, a simple result calculator using the WAEC grade bands, a digital version of the Week 9 tally survey.
- Judges look for a real use, a working demo (even small), and an honest limitation. A small finished thing beats a big unfinished one.

7. Certificates and Awards

- Every participant receives a printed Certificate of Participation on the day.
- First, second and third place teams receive winner certificates. Print one per pupil, with the team's school name entered by hand.

- The printable PDF certificates ship with this pack: blank name and school lines, the Rehoteq purple and hexagon theme, and a date line for the host to fill. They print on A4 landscape.
- Suggested prizes the host may add: exercise books, stationery sets, data-enabled time in the school computer room, or a small trophy for the winning school to keep until the next event.

8. Host School Checklist

18. Confirm the date with all participating schools at least 14 days ahead, and send this pack with the invitation.
19. Prepare the hall or rooms: one table per team, one computer per team where available (a shared machine with Scratch, a browser and Python installed), and paper answer sheets.
20. Print: quiz papers and answer sheets, challenge briefs per level, rubric sheets per judge, certificates (participation for every registered pupil, plus winner sets).
21. Brief the judges on the rubric before the event and run one calibration example together.
22. Plan the unplugged fallbacks so the day works even if computers fail.
23. Keep water, first aid and a quiet corner available, and hold the signed consent records at the registration desk.
24. After the event: post scores to each school within 48 hours, and invite the winning school to host the next edition.

9. Document Status

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